CHAPTER 13

PROBABILITY

CHAPTER ESSENCE AT A GLANCE:
Conditional probability, multiplication theorem, law of total probability, Bayes' theorem, and independent events.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Conditional Probability and Sample Space Reduction:** The conditional probability of event A given event B restricts the sample space to event B, calculated as the probability of their intersection divided by the probability of B.

• **Independent Events and Multiplication:** Two events are independent when the occurrence of one does not alter the likelihood of the other, which occurs if and only if their joint probability equals the product of their individual probabilities.

• **Total Probability Across Partitions:** The total probability of an outcome is the weighted sum of its conditional probabilities evaluated across a mutually exclusive and exhaustive partition of the sample space.

• **Bayes Theorem and Reverse Inference:** Inverts conditional probabilities to compute the posterior probability that a specific observed outcome arose from a particular causative partition hypothesis.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0)$$

Definition of conditional probability of event A given event B

$$P(A \cap B) = P(A) \cdot P(B) \Leftrightarrow A \text{ and } B \text{ are independent}$$

Multiplication rule for statistically independent events

$$P(A) = \sum_{i=1}^{n} P(E_i) \cdot P(A \mid E_i)$$

Theorem of Total Probability across partition events

$$P(E_i \mid A) = \frac{P(E_i) \cdot P(A \mid E_i)}{\sum_{j=1}^{n} P(E_j) \cdot P(A \mid E_j)}$$

Bayes Theorem for posterior probability of causative event

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Addition theorem for probability of union of two events

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Bayes Event Definition] In Bayes' theorem, define events E_1, E_2... and condition A in explicit English sentences before writing probabilities! Evaluators deduct 1 Mark if events are not declared.

• **Alert:** Ensure boxed final results clearly state mathematical units or interval notation.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define conditional probability P(A|B). State formula and properties. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Conditional Probability Definition]** If A and B are two events associated with the same sample space S of a random experiment, then the conditional probability of occurrence of event A given that event B has already occurred (with $P(B) \neq 0$) is: $P(A \mid B) = P(A \cap B)$ over $P(B)$. [1 Mark]

• **[Key Properties]** 1. $P(S \mid B) = P(B \mid B) = 1$. 2. $P(A' \mid B) = 1 - P(A \mid B)$. 3. $P((A \cup B_1) \mid B) = P(A \mid B) + P(B_1 \mid B) - P((A \cap B_1) \mid B)$. [1 Mark]

• **[Boxed Result: $P(A \mid B) = P(A \cap B)$ over $P(B)$; $P(A' \mid B) = 1 - P(A \mid B)$]**

2. If P(A) = 0.6, P(B) = 0.3, and P(A B) = 0.2, find P(A|B) and P(B|A). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula]** $P(A \mid B) = P(A \cap B)$ over $P(B)$ and $P(B \mid A) = P(A \cap B)$ over $P(A)$. [0.5 Mark]

• **[Step 2: Calculations]** Given $P(A) = 0.6$, $P(B) = 0.3$, $P(A \cap B) = 0.2$. $P(A \mid B) = 0.2$ over $0.3 = 2$ over $3 \approx 0.667$. [0.5 Mark]

• $P(B \mid A) = 0.2$ over $0.6 = 2$ over $6 = 1$ over 3 approx. 0.333. [1 Mark]

• **[Boxed Result: $P(A \mid B) = 2$ over 3 and $P(B \mid A) = 1$ over 3]**

3. If P(A) = 0.8, P(B) = 0.5, and P(B|A) = 0.4, find P(A B), P(A|B), and P(A B). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Given Data]** $P(A) = 0.8$, $P(B) = 0.5$, $P(B \mid A) = 0.4$. [0.5 Mark]

• **[Calculations]** 1. $P(A \cap B) = P(A) \cdot P(B \mid A) = 0.8 \times 0.4 = 0.32$. [1 Mark]

• 2. $P(A \mid B) = P(A B)$ over $P(B) = 0.32$ over $0.5 = 32$ over $50 = 16$ over $25 = 0.64$. [0.5 Mark]

• 3. $P(A B) = P(A) + P(B) - P(A B) = 0.8 + 0.5 - 0.32 = 1.3 - 0.32 = 0.98$. [1 Mark]

• **[Boxed Result: $P(A \cap B) = 0.32$; $P(A \mid B) = 0.64$; $P(A \cup B) = 0.98$]**

4. A die is thrown three times. Events: A = 4 appears on 3rd toss, B = 6 and 5 appear respectively on 1st two tosses. Find P(A|B). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Sample Spaces of Throws]** Total outcomes = $6^3 = 216$. Event B: 6 and 5 appear respectively on first two tosses. B = {(6, 5, 1), (6, 5, 2), (6, 5, 3), (6, 5, 4), (6, 5, 5), (6, 5, 6)}. Total n(B) = 6. [1 Mark]

• Event A: 4 appears on 3rd toss. A B = {(6, 5, 4)}. Total n(A B) = 1. [1 Mark]

• $P(A \mid B) = n(A B)$ over $n(B) = 1$ over 6. [1 Mark]

• **[Boxed Result: $P(A \mid B) = 1$ over 6]**

5. Define independent events. If A and B are independent, prove that A and B' are also independent. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Independent Events Definition]** Two events A and B are independent if $P(A \cap B) = P(A) \cdot P(B)$. [1 Mark]

• **[Proof that A and B' are Independent]** Notice: $A = (A \cap B) \cup (A \cap B')$ where $(A \cap B)$ and $(A \cap B')$ are mutually exclusive. $P(A) = P(A \cap B) + P(A \cap B')$. $P(A \cap B') = P(A) - P(A \cap B)$. Since A and B are independent, $P(A \cap B) = P(A) P(B)$: $P(A \cap B') = P(A) - P(A) P(B) = P(A) [1 - P(B)] = P(A) \cdot P(B')$. Hence A and B' are independent events! Proved. [2 Marks]

• **[Boxed Result: $P(A \cap B') = P(A) \cdot P(B') \Rightarrow$ A and B' are independent]**

6. Bag I contains 3 red and 4 black balls, Bag II contains 5 red and 6 black balls. One ball is transferred from Bag I to Bag II and then a ball is drawn from Bag II. The ball drawn is red. Find probability that transferred ball was black. [Score: 5 Marks]

Answer & Key Valuation Steps:

• **[Events Setup]** Bag I: 3 Red, 4 Black (Total 7). Bag II: 5 Red, 6 Black (Total 11). E_1: Transferred ball is RED $\Rightarrow$ $P(E_1) = 3$ over 7. E_2: Transferred ball is BLACK $\Rightarrow$ $P(E_2) = 4$ over 7. A: Ball drawn from Bag II is RED. [1.5 Marks]

• **[Conditional Probabilities]** If Red transferred, Bag II has 6 Red, 6 Black: $P(A \mid E_1) = 6$ over 12 = 1 over 2. If Black transferred, Bag II has 5 Red, 7 Black: $P(A \mid E_2) = 5$ over 12. [1.5 Marks]

• **[Bayes' Theorem for $P(E_2 \mid A)$]** $P(E_2 \mid A) = [\, P(E_2)\, P(A \mid E_2) \,]$ over $[\, P(E_1)\, P(A \mid E_1) + P(E_2)\, P(A \mid E_2) \,] = [$ (4 over 7) × (5 over 12)] over [(3 over 7) × (6 over 12) + (4 over 7) × (5 over 12)] = (20 over 84) over [(18 + 20) over 84] = 20 over 38 = 10 over 19. [2 Marks]

• **[Boxed Result: P(transferred ball was black | red drawn) = 10 over 19]**

7. In a factory, machines A, B, C manufacture 25%, 35%, 40% of bolts. Defective percentages are 5%, 4%, 2%. A bolt drawn is defective. Find probability it was manufactured by machine B. [Score: 5 Marks]

Answer & Key Valuation Steps:

• **[Events Setup]** E_1, E_2, E_3: Bolt manufactured by machines A, B, C. $P(E_1) = 0.25$, $P(E_2) = 0.35$, $P(E_3) = 0.40$. D: Drawn bolt is defective. $P(D \mid E_1) = 0.05$, $P(D \mid E_2) = 0.04$, $P(D \mid E_3) = 0.02$. [1.5 Marks]

• **[Bayes' Theorem for $P(E_2 \mid D)$]** $P(E_2 \mid D) = [\, P(E_2)\, P(D \mid E_2) \,]$ over $[\, P(E_1)\, P(D \mid E_1) + P(E_2)\, P(D \mid E_2) + P(E_3)\, P(D \mid E_3) \,]$ Numerator: 0.35 × 0.04 = 0.0140. Denominator: (0.25 × 0.05) + (0.35 × 0.04) + (0.40 × 0.02) = 0.0125 + 0.0140 + 0.0080 = 0.0345. $P(E_2 \mid D)$ = 0.0140 over 0.0345 = 140 over 345 = 28 over 69 ≈ 0.406. [3.5 Marks]

• **[Boxed Result: P(Machine B | Defective) = 28 over 69 ≈ 0.406]**